

3E3 Modelling Risk Coursework

GreenCo Energy is developing a coastal renewable energy hub integrating offshore wind generation, on-site hydrogen production, and a refuelling station serving heavy goods vehicles (HGVs). The project is strategically important: management sees it as a first-mover opportunity in zero-carbon freight corridors. However, there is significant uncertainty around demand growth, operational performance, congestion risk, and capital commitment.

Hydrogen demand from fleet operators follows a normal distribution with a mean of 1,000 tonnes per month and a standard deviation of 150 tonnes. Assume monthly demand is independently distributed and that production decisions are made at the beginning of each month before demand is realised. Hydrogen sells for £4,000 per tonne and costs £2,500 per tonne to produce. If production falls short, hydrogen must be purchased on the spot market at £3,200 per tonne to meet contractual obligations. Unsold hydrogen incurs £300 per tonne in storage and handling costs and cannot be carried forward beyond one month.

The electrolyser has a nominal capacity of 40 tonnes per day, with operational availability of 85 percent due to maintenance and wind intermittency. Assume 30 days per month.

At the refuelling station, there are initially three refuelling bays operating in parallel. HGVs arrive according to a Poisson process at an average rate of 10 per hour. Refuelling times are exponentially distributed with a mean of 12 minutes per vehicle. Fleet contracts require that the average waiting time in queue remains below 5 minutes.

GreenCo must now decide how to scale its hydrogen infrastructure over a four-year horizon. Assume all cash flows occur annually and are discounted at 7 percent per year. Demand growth remains uncertain and follows one of two trajectories. Growth affects the mean demand, while the coefficient of variation remains constant over time. High growth (probability 0.45): mean demand increases by 10 percent per year. Low growth (probability 0.55): mean demand increases by 3 percent per year. In the high-growth scenario, there is a 40 percent probability that a competing hydrogen provider enters the market in Year 3, reducing realised demand from that point onward by 15 percent permanently.

GreenCo has to choose one strategy from the following options:

Strategy 1: Immediate Scale-Up

GreenCo expands electrolyser capacity from 40 to 65 tonnes per day immediately. This requires an upfront capital investment of £48 million in Year 0. Operational availability improves to 95 percent due

to system redundancy. The expansion also requires installing two additional refuelling bays at a cost of £6 million. If high growth materialises, the expanded system allows GreenCo to capture 100 percent of realised demand subject to operational availability. If low growth materialises, capacity utilisation falls below 65 percent from Year 2 onward, increasing unit operating cost by £150 per tonne due to scale inefficiencies.

Strategy 2: Flexibility and Risk Mitigation

GreenCo invests £20 million in Year 0 in modular storage, demand-smoothing contracts, and digital forecasting systems. This reduces demand standard deviation by 40 percent at all future periods and lowers emergency procurement cost from £3,200 to £2,900 per tonne through pre-negotiated contingent contracts. No additional production capacity is installed. Under high growth, expected demand exceeds effective production capacity from Year 3 onward, leading to systematic reliance on emergency procurement. Under low growth, volatility reduction materially improves profit stability and reduces downside risk exposure.

Strategy 3: Staged Expansion

GreenCo invests £8 million in Year 0 to install expandable foundations and grid upgrades that preserve the option to scale capacity later at lower cost. At the end of Year 1, after observing realised demand growth: If high growth is observed, GreenCo can expand capacity to 65 tonnes per day at a reduced cost of £42 million (instead of £48 million). If low growth is observed, GreenCo can instead invest £15 million in flexibility measures (a reduced version of Strategy 2). However, delaying full commitment reduces realised revenue by 10 percent in Years 1 and 2 under both growth scenarios.

Management has expressed concerns regarding demand volatility, production variability, congestion risk, and long-term investment decisions under conditions of uncertainty. Drawing on your expertise, you are required to develop an integrated risk analysis that supports strategic and operational decision-making. The objective is to maximise expected net present value while explicitly assessing downside risk exposure and service-level compliance. Your report should systematically identify and evaluate the key sources of business risk and the associated operational and financial challenges. It must clearly explain the analytical framework, assumptions (particularly in the absence of any key information), and methods used, ensuring transparency in how conclusions are derived. In addition to diagnosing risks, the report should provide well-justified managerial recommendations that enhance resilience, improve performance stability, and support informed long-term investment decisions under uncertainty. The board has limited availability; therefore, your report must not exceed 8 pages.